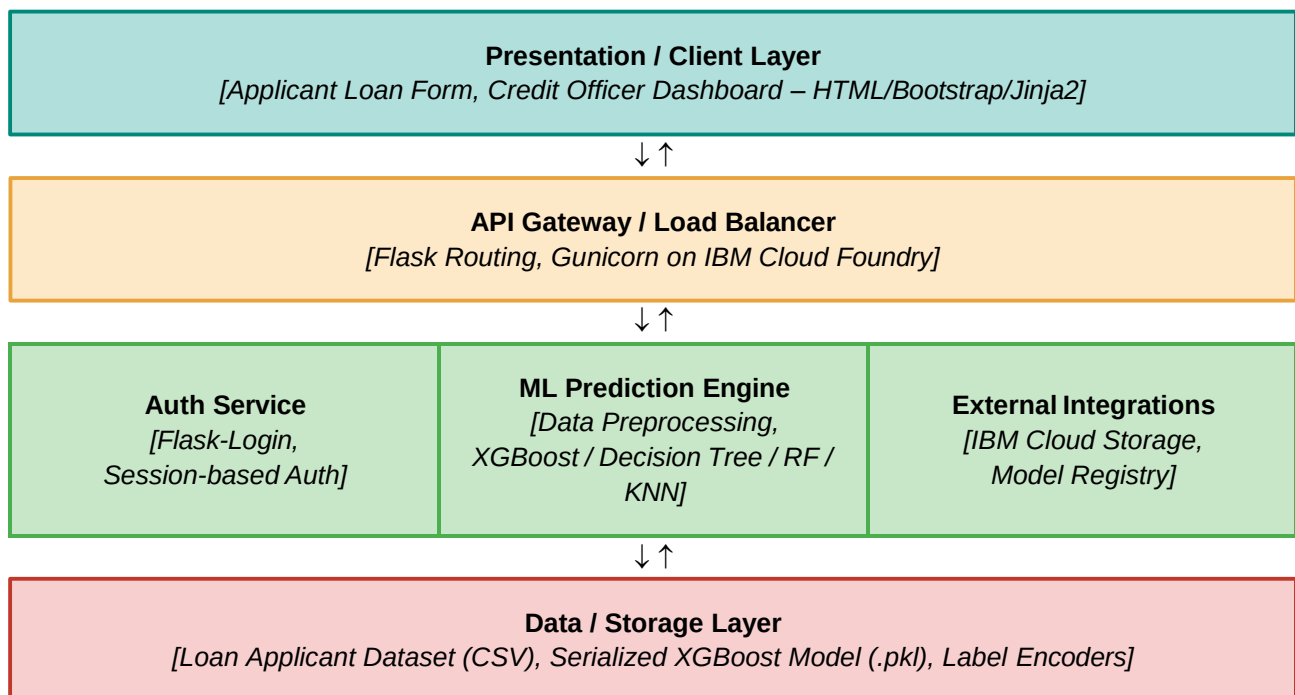


## Solution Architecture

|                      |              |
|----------------------|--------------|
| <b>Date</b>          | 06 July 2026 |
| <b>Team ID</b>       | xxxxxx       |
| <b>Project Name</b>  | Smart Lender |
| <b>Maximum Marks</b> | 5 Marks      |

### Solution Architecture Diagram:

Below is the high-level visual representation of the solution's architecture, mapped out across the Presentation, Application, and Data layers.



## Component Description Table

| Component Name     | Description / Role in Architecture                                                                                                                                                                                                                                                | Technologies Used                            |
|--------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------|
| Presentation Layer | Acts as the user-facing interface. It captures applicant inputs (gender, marital status, education, employment status, income, loan amount, loan term, credit history, property area), displays the predicted approval outcome, and renders result summaries for credit officers. | HTML5, CSS3, Bootstrap, JavaScript, Jinja2   |
| API Gateway        | Serves as the entry point for all client requests. It routes HTTP requests to the appropriate backend logic, handles URL routing, manages sessions, and returns formatted prediction responses to the frontend.                                                                   | Flask Framework, Unicorn                     |
| Microservices      | Contains the core business logic and processing engines. Includes the ML Engine (data preprocessing and XGBoost inference), the Model Comparison Module (Decision Tree, Random Forest, KNN, XGBoost evaluation), and the Auth Module for secure officer login.                    | Python, Scikit-learn, XGBoost, Pandas, NumPy |
| Database           | Persists structured data required for the application's operation. Stores the serialized trained model, encoders for categorical features, and logs of past applicant predictions for audit and batch review.                                                                     | Pickle / Joblib, SQLite                      |